

Electrical Machines

Previous Year Questions • 2013–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electrical Machines (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:



 Easy  Medium  Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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Contents

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UNIT

I

Magnetic Circuits

Topics

1. Magnetically Coupled Circuits, Dot Convention
2. Energy in Coupled circuits
3. Electric and Magnetic Materials

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Unit Snapshot*

Stream: IN	**Questions:** 6	**MCQ:** 4	**MSQ:** 0	**NAT:** 2
1M: 2	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2025)–4M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MCQ 2025 1M

The circuit shown in Figure (a) can be represented using its equivalent T-model as shown in Figure (b). The values of the inductances L_1, L_2, and L_3 in the equivalent T-model are

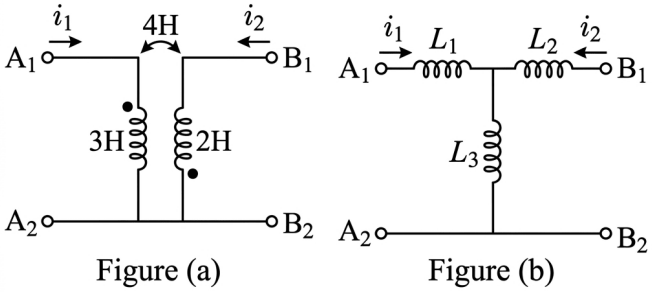


Fig. Q1.25.1

A. $L_1 = 7\,\text{H}, L_2 = 6\,\text{H}, L_3 = -4\,\text{H}$

B. $L_1 = -1\,\text{H}, L_2 = -2\,\text{H}, L_3 = 4\,\text{H}$

C. $L_1 = 3\,\text{H}, L_2 = 2\,\text{H}, L_3 = 9\,\text{H}$

D. $L_1 = 1\,\text{H}, L_2 = -2\,\text{H}, L_3 = -4\,\text{H}$

Q1.24.1 NAT 2024 2M

Two magnetically coupled coils, when connected in series-aiding configuration, have a total inductance of 500 mH. When connected in series-opposing configuration, the coils have a total inductance of 300 mH. If the self-inductance of both the coils are equal, then the coupling coefficient is _____ (rounded off to two decimal places).

Q1.23.1 NAT 2023 2M

A sinusoidal current of $i_1(t) = 1\sin(200\pi t)$ mA is flowing through a 4 H inductor which is mutually coupled to another 5H inductor carrying
$i_2(t) = 2\sin(200\pi t)$ mA as shown in the figure. The coupling coefficient between the inductors is 0.6. The peak energy stored in the circuit is _____ μJ (rounded off to two decimal places).

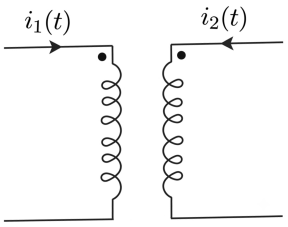


Fig. Q1.23.1

Q1.15.1 MCQ 2015 1M

The output voltage of the ideal transformer with the polarities and dots shown in the figure is given by

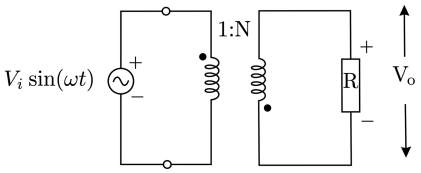


Fig. Q1.15.1

A. $NV_i \sin \omega t$

B. $-NV_i \sin \omega t$

C. $\frac{1}{N}V_i \sin \omega t$

D. $-\frac{1}{N}V_i \sin \omega t$

Q1.13.1 MCQ 2013 2M

The following arrangement consists of an ideal transformer and an attenuator, which attenuates by a factor of 0.8 An ac voltage $V_{WX1} = 100\,\text{V}$ is applied across WX to get an open circuit voltages V_{YZ1} across YZ Next, an ac voltage $V_{YZ2} = 100\,\text{V}$ is applied across YZ to get an open circuit voltage V_{wx2} across WX Then, V_{YZ1}/V_{WX1}, V_{WX2}/V_{YZ2} are respectively,

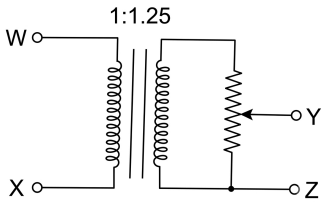


Fig. Q1.13.1

A. 125/100 and 80/100

B. 100/100 and 80/100

C. 100/100 and 100/100

D. 80/100 and 80/100

Q1.13.2 MCQ 2013 2M

Two magnetically uncoupled inductive coils have Q factors q_1 and q_2 at the chosen operating frequency. Their respective resistances are R_1 and R_2 When connected in series, the effective Q factor of the series combination at the same operating frequency is

A. $q_1 + q_2$

B. $(1/q_1) + (1/q_2)$

C. $(q_1R_1 + q_2R_2)/(R_1 + R_2)$

D. $(q_1R_2 + q_2R_1)/(R_1 + R_2)$

Answer Key

Q1.25.1 **A**	Q1.24.1 **0.25**	Q1.23.1 **17.37**
Q1.15.1 **B**	Q1.13.1 **B**	Q1.13.2 **C**

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UNIT

II

Single Phase Transformers

Topics

1. Equivalent Circuit and Phasor Diagram
2. Open Circuit and Short Circuit Tests
3. Voltage Regulation and Efficiency of Transformer

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Unit Snapshot*

Stream: IN	**Questions:** 7	**MCQ:** 2	**MSQ:** 0	**NAT:** 5
1M: 2	**2M:** 5	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2025)–3M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

A single phase step-down transformer has a high voltage primary winding and a low voltage secondary winding. The secondary winding is connected to a load impedance Z_2. The impedance Z_2 when referred to the primary winding is denoted by Z_2'. Which of the following conditions is true?

A. $Z_2' = Z_2$ **B.** $0.5Z_2 < Z_2' < Z_2$
C. $Z_2' > Z_2$ **D.** $Z_2' \leq 0.5Z_2$

Q2.26.2 NAT 2026 2M

A single phase transformer has a rating of 10kVA, 50Hz, 1100V/220V. In an open circuit test with the high voltage side open, the following are obtained:

- Voltage applied to the low voltage winding = 220V
- Measured power = 363W
- Measured current = 2.75A

Magnetizing reactance X_m referred to the high voltage winding is _____ Ω.

Q2.25.1 NAT 2025 2M

The circuit given in the figure is driven by a voltage source $V_s = 25\sqrt{2}\angle 30°$ V. The system is operating at a frequency of 50 Hz. The transformers are assumed to be ideal. The average power dissipated, in W, in the 50 kΩ resistance is _____ (rounded off to two decimal places).

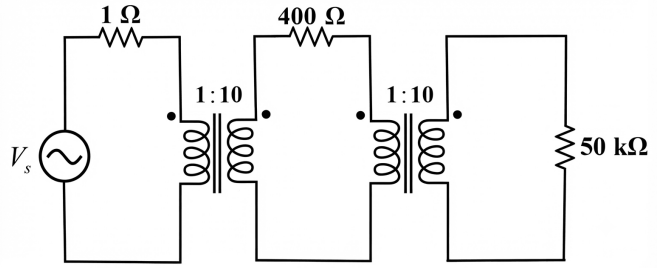


Fig. Q2.25.1

Q2.24.1 NAT 2024 2M

A 50 kVA transformer has an efficiency of 95% at full load and unity power factor. Assume the core losses are negligible. The efficiency of the transformer at 75% of the full load and 0.8 power factor is _____ % (rounded off to one decimal place).

Q2.23.1 NAT 2023 2M

A short circuit test is conducted on a single-phase transformer by shorting its secondary. The frequency of input voltage is 1kHz. The corresponding wattmeter reading, primary current and primary voltage are 8 W, 2 A and 6 V respectively. Assume that the no-load losses and the no-load currents are negligible, and the core has linear magnetic characteristics. Keeping the secondary shorted, the primary is connected to a 2 V (RMS), 1 kHz sinusoidal source in series with a $\frac{1}{2\pi\sqrt{5}}$ mF capacitor. The primary current (RMS) will be _____ A (rounded off to two decimal places).

Q2.21.1 NAT 2021 1M

A single-phase transformer has a magnetizing inductance of 250 mH and a core loss resistance of 300 Ω, referred to primary side. When excited with a 230 V, 50 Hz sinusoidal supply at the primary, the power factor of the input current drawn, with secondary on open circuit, is _____ (rounded off to two decimal places).

Q2.21.2 MCQ 2021 2M

A single-phase transformer has maximum efficiency of 98%. The core losses are 80 W and the equivalent winding resistance as seen from the primary side is 0.5 Ω. The rated current on the primary side is 25 A. The percentage of the rated input current at which the maximum efficiency occurs is

A. 50.6% **B.** 80.5%
C. 100% **D.** 35.7%

Answer Key

Q2.26.1 C	Q2.26.2 2490 to 2510	Q2.25.1 31.00 to 31.50
Q2.24.1 95.2 to 95.4	Q2.23.1 1	Q2.21.1 0.24 to 0.26
Q2.21.2 A		

UNIT

III

Three Phase Induction Machines

Topics

1. Principle of Operation , Types and Performance
2. Equivalent Circuit of Induction Motor
3. Torque-Slip, Torque-Speed Characteristics
4. Stability and Testing – No-Load (OC) and Blocked-Rotor (SC) Tests
5. Starting of Induction Motors
6. Speed Control
7. Types of losses and efficiency calculation of Electric Machines

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Unit Snapshot*

Stream: IN	**Questions:** 5	**MCQ:** 2	**MSQ:** 0	**NAT:** 3
1M: 2	**2M:** 3	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 MCQ 2025 2M

A 2-pole, 50 Hz, 3-phase induction motor supplies power to a certain load at 2970 rpm. The torque-speed curve of this machine follows a linear relationship between synchronous speed and 95 % of synchronous speed. Assume mechanical and stray losses to be zero. If the load torque of the motor is doubled, the new operating speed of the motor, in rpm, is

A. 2940 **B.** 2812
C. 2970 **D.** 2850

Q3.24.1 NAT 2024 2M

A three-phase squirrel-cage induction motor has a starting torque of 100 % of the full load torque and a maximum torque of 300 % of the full load torque. Neglecting the stator impedance, the slip at the maximum torque is _____ % (rounded off to two decimal places).

Q3.23.1 NAT 2023 1M

The no-load steady-state output voltage of a DC shunt generator is 200 V when it is driven in the clockwise direction at its rated speed. If the same machine is driven at the rated speed but in the opposite direction, the steady-state output voltage will be _____ V (rounded off to the nearest integer).

Q3.22.1 NAT 2022 1M

A 440 V, 8 kW, 4-pole, 50 Hz, star-connected induction motor has a full load slip of 0.04. The rotor speed (in rpm) at full load is _____ (round off to one decimal place)

Q3.21.1 MCQ 2021 2M

A slip-ring induction motor is expected to be started by adding extra resistance in the rotor circuit. The benefit that is derived by adding extra resistance in the rotor circuit in comparison to the rotor being shorted is

A. The starting torque would be higher.

B. The losses at starting would be lower.

C. The starting current is higher.

D. The power factor at start will be lower.

Answer Key

Q3.25.1 A	Q3.24.1 17.00 to 17.30	Q3.23.1 0
Q3.22.1 1440	Q3.21.1 A	

Also in this Series

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